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MOWER WITH MOTOR BRACKETS

Abstract

A mower includes a chassis, a tractive element coupled to the chassis, a cutting unit coupled to the chassis and including a cutting element, a mounting bracket coupled to the chassis, and a motor configured to drive the cutting element. The motor is configured to be removably coupled to the cutting unit and configured to be removably coupled to the mounting bracket. The mower is reconfigurable between (a) an operating configuration in which the motor is removed from the mounting bracket and coupled to the cutting unit and (b) a service or transportation configuration in which the motor is removed from the cutting unit and coupled to the mounting bracket.

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Background/Summary

BACKGROUND

[0001] The present disclosure relates generally to mowers. More specifically, the present disclosure relates to a cutting unit for a mower.

[0002] Mowers are used to maintain vegetation (e.g., grass, clover, weeds, etc.) at a desired height. To accomplish this, mowers include at least one cutting unit having a cutting element that is driven by a motor. When performing maintenance or during transportation, the cutting units may be removed. However, the motors may remain coupled to a chassis of the mower (e.g., by one or more cables), even after the other portions of the cutting units are removed. Permitting these motors to hang freely from the chassis may cause them to become tangled or otherwise become damaged.

SUMMARY

[0003] One embodiment relates to a mower. The mower includes a chassis, a tractive element coupled to the chassis, a cutting unit removably coupled to the chassis and including a cutting element, a mounting bracket coupled to the chassis, and a motor configured to drive the cutting element. The motor is configured to be removably coupled to the cutting unit and configured to be removably coupled to the mounting bracket. The mower is reconfigurable between (a) an operating configuration in which the motor is removed from the mounting bracket and coupled to the cutting unit and (b) a service or transportation configuration in which the motor is removed from the cutting unit and coupled to the mounting bracket.

[0004] Another embodiment relates to a mower. The mower includes a chassis, a tractive element coupled to the chassis, a cutting unit coupled to the chassis and including a cutting element, an actuator configured to raise the cutting unit relative to the chassis, a mounting bracket coupled to the chassis, the mounting bracket defining a first fastener passage and a second fastener passage, a motor configured to drive the cutting element, the motor defining a first fastener aperture and a second fastener aperture, a first fastener, and a second fastener. The mower is reconfigurable between a first configuration and a second configuration. In the first configuration, (a) the first fastener extends through the first fastener aperture and engages the cutting unit and (b) the second fastener extends through the second fastener aperture and engages the cutting unit. In the second configuration, (a) the first fastener extends through the first fastener aperture and into the first fastener passage to couple the motor to the mounting bracket and (b) the second fastener extends into the second fastener passage without extending through the second fastener aperture.

[0005] Still another embodiment relates to a mower. The mower includes a chassis, a tractive element coupled to the chassis, a first cutting unit configured to be driven by a first motor, a second cutting unit configured to be driven by a second motor, a first mounting bracket pivotally coupled to the chassis, and a second mounting bracket fixedly coupled to the chassis. The mower is reconfigurable between a first configuration and a second configuration. In the first configuration, (a) the first motor is coupled to the first cutting unit and (b) the second motor is coupled to the second cutting unit. In the second configuration, (a) the first motor is disconnected from the first cutting unit and coupled to the first mounting bracket and (b) the second motor is disconnected from the second cutting unit and coupled to the second mounting bracket.

[0006] This summary is illustrative only and is not intended to be in any way limiting. Other aspects, inventive features, and advantages of the devices or processes described herein will become apparent in the detailed description set forth herein, taken in conjunction with the accompanying figures, wherein like reference numerals refer to like elements.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. 1 is a perspective view of a mower, according to an exemplary embodiment.

[0008] FIG. 2 is a schematic block diagram of the mower of FIG. 1, according to an exemplary embodiment.

[0009] FIG. 3 is a perspective view of the mower of FIG. 1 in a service configuration.

[0010] FIG. 4 is a front perspective view of a motor coupled to a front cutting unit of the mower of FIG. 1.

[0011] FIG. 5 is a front perspective view of a lift arm for the front cutting unit of FIG. 4.

[0012] FIG. 6 is a rear view of the lift arm of FIG. 5.

[0013] FIG. 7 is a rear perspective view of the lift arm of FIG. 5 installed on the mower of FIG. 1.

[0014] FIG. 8 is a front perspective view of the mower of FIG. 1 with the front cutting unit of FIG. 4 removed and the motor of FIG. 4 hanging freely.

[0015] FIG. 9 is a front perspective view of the motor of FIG. 4 and the lift arm of FIG. 5 being assembled into the service configuration.

[0016] FIG. 10 is a front perspective view of the motor of FIG. 4 and the lift arm of FIG. 5 assembled into the service configuration.

[0017] FIG. 11 is a right side view of the motor of FIG. 4 and the lift arm of FIG. 5 assembled into the service configuration.

[0018] FIG. 12A is a front perspective view of a rear cutting unit of the mower of FIG. 1.

[0019] FIG. 12B is a rear perspective view of a lift arm and a motor for use with the rear cutting unit of FIG. 12A.

[0020] FIG. 13 is a front perspective view of a motor coupled to a central cutting unit of the mower of FIG. 1.

[0021] FIG. 14 is a rear perspective view of a mounting bracket for the motor of FIG. 13.

[0022] FIG. 15 is a rear perspective view of the motor of FIG. 13 and the mounting bracket of FIG. 14 being assembled into the service configuration.

[0023] FIG. 16 is a top perspective view of the motor of FIG. 13 and the mounting bracket of FIG. 14 assembled into the service configuration.

[0024] FIG. 17 is a bottom perspective view of the motor of FIG. 13 and the mounting bracket of FIG. 14 assembled into the service configuration.

DETAILED DESCRIPTION

[0025] Before turning to the figures, which illustrate certain exemplary embodiments in detail, it should be understood that the present disclosure is not limited to the details or methodology set forth in the description or illustrated in the figures. It should also be understood that the terminology used herein is for the purpose of description only and should not be regarded as limiting.

[0026] According to an exemplary embodiment, a mower of the present disclosure includes a series of mower decks, each driven by a motor. When transporting the mower or servicing the mower, it may be advantageous to remove the mower decks (e.g., to provide clearance around certain components). However, after the mower decks are removed, the motors remain coupled to the chassis of the mower. If left hanging freely from the cables, the motors would be susceptible to damage. Instead, the mower of the exemplary embodiment includes a series of mounting brackets, to which the motors are attached. This limits the movement of the motors, preventing the cables from becoming tangled or other damage to the motors. Some of the mounting brackets are built into lift arms for the mower decks, and another mounting bracket is fixed to the chassis. Each of the mounting brackets permits the same fasteners that secured the motors to the mower decks to also be used to secure the motors to the mounting brackets.

Overall Vehicle

[0027] As shown in FIGS. 1-3, a machine or vehicle, shown as mower 10, includes a chassis, shown as frame 12; a body assembly, shown as body 20, coupled to the frame 12 and having an occupant portion or section, shown as occupant seating area 30; operator input and output devices, shown as operator controls 40, that are disposed within the occupant seating area 30; a drivetrain,

shown as driveline **50**, coupled to the frame **12** and at least partially disposed under the body **20**; a vehicle suspension system, shown as suspension system **60**, coupled to the frame **12** and one or more components of the driveline **50**; a vehicle braking system, shown as braking system **70**, coupled to one or more components of the driveline **50** to facilitate selectively braking the one or more components of the driveline **50**; and a vehicle control system, shown as control system **100**, coupled to the operator controls **40**, the driveline **50**, the suspension system **60**, the braking system **70**, and/or one or more external devices or systems positioned remote from the mower **10**, shown as remote system(s) **200**. The mower **10** further includes a series of implements, mower assemblies, or cutting units, shown as mower decks **80**. In other embodiments, the mower **10** includes more or fewer components.

[0028] According to an exemplary embodiment, the mower **10** is an off-road machine or vehicle. In some embodiments, the mower **10** is a lawnmower, a turf mower, a push mower, a ride-on mower, a stand-on mower, or another type of mower. In other embodiments, the off-road machine or vehicle is a lightweight or recreational machine or vehicle such as a golf cart, an all-terrain vehicle (“ATV”), a utility task vehicle (“UTV”), and/or another type of lightweight or recreational machine or vehicle. In some embodiments, the off-road machine or vehicle is a chore product such as aerator, turf sprayer, bunker rake, and/or another type of chore product (e.g., that may be used on a golf course).

[0029] According to the exemplary embodiment shown in FIG. **1**, the occupant seating area **30** includes a single seat, shown as driver seat **32**. In some embodiments, the occupant seating area **30** includes additional seats (e.g., a passenger seat, an additional row of seats, etc.). According to the exemplary embodiment shown in FIG. **1**, the driver seat **32** is laterally centered on the body **20** and facing forward. In some embodiments, the driver seat **32** is facing rearward or otherwise positioned. In some embodiments, the occupant seating area **30** is omitted (e.g., the mower **10** is configured as a push mower). A portion of the frame **12** defines a platform, deck, or standing area, shown as operator platform **34**. The operator platform **34** may extend forward of the driver seat **32** such that the occupant can rest their feet on the operator platform **34** while seated in the driver seat **32**. The operator platform **34** may support the occupant as the occupant enters or exits the driver seat **32**.

[0030] According to an exemplary embodiment, the operator controls **40** are configured to provide an operator with the ability to control one or more functions of and/or provide commands to the mower **10** and the components thereof (e.g., turn on, turn off, drive, turn, brake, engage various operating modes, raise/lower a mower deck **80**, etc.). As shown in FIGS. **1** and **2**, the operator controls **40** include a steering interface (e.g., a steering wheel, joystick(s), etc.), shown as steering wheel **42**, an accelerator interface and/or braking interface (e.g., a pedal, a throttle, etc.), shown as traction pedal **44**, and one or more additional interfaces, shown as operator interface **48**. The traction pedal **44** may be used to control the speed and direction of travel of the mower **10**. By way of example, pressing the traction pedal **44** in a first direction may cause the driveline **50** to move the mower **10** forward, and pressing the traction pedal **44** in an opposing section direction may cause the driveline **50** to move the mower **10** rearward. Returning the traction pedal **44** to a middle or neutral position may cause the braking system **70** and/or the driveline **50** to slow or stop the mower **10** or to hold the mower **10** in place. Alternatively, the operator interface **48** may include a pair of handles that control the driveline **50** in a zero-turn configuration (e.g., a left joystick to control the left side of the driveline **50** and a right joystick to control a right side of the driveline **50**). The operator interface **48** may be used to control operation of the mower decks **80** (e.g., changing a cutting speed of a mower deck **80**, changing a cutting height of a mower deck **80**, etc.). The operator interface **48** may include one or more displays and one or more input devices. The one or more displays may be or include a touchscreen, an LCD display, a LED display, a speedometer, gauges, warning lights, etc. The one or more input device may be or include buttons, switches, knobs, levers, dials, etc.

[0031] According to an exemplary embodiment, the driveline **50** is configured to propel the mower **10**. As shown in FIGS. **1** and **2**, the driveline **50** includes a primary driver, shown as prime mover **52**, an energy storage device, shown as energy storage **54**, a first tractive assembly (e.g., axles, wheels, tracks, differentials, etc.), shown as rear tractive assembly **56**, and a second tractive assembly (e.g., axles, wheels, tracks, differentials, etc.), shown as front tractive assembly **58**. In some embodiments, the driveline **50** is a conventional driveline whereby the prime mover **52** is an internal combustion engine and the energy storage **54** is a fuel tank. The internal combustion engine may be a spark-ignition internal combustion engine or a compression-ignition internal combustion engine that may use any suitable fuel type (e.g., diesel, ethanol, gasoline, natural gas, propane, etc.). In some embodiments, the driveline **50** is an electric driveline whereby the prime mover **52** is one or more electric motors and the energy storage **54** is a battery system. In some embodiments, the driveline **50** is a fuel cell electric driveline whereby the prime mover **52** is one or more electric motors and the energy storage **54** is a fuel cell (e.g., that stores hydrogen, that produces electricity from the hydrogen, etc.). In some embodiments, the driveline **50** is a hybrid driveline whereby (i) the prime mover **52** includes an internal combustion engine and an electric motor/generator and (ii) the energy storage **54** includes a fuel tank and/or a battery system. According to the exemplary embodiment shown in FIG. **1**, the rear tractive assembly **56** includes rear tractive elements and the front tractive assembly **58** includes front tractive elements that are configured as wheels. In some embodiments, the rear tractive elements and/or the front tractive elements are configured as tracks. In some embodiments, the driveline **50** is omitted, and the mower **10** is propelled by an operator (e.g., the mower **10** is configured as a push mower).

[0032] According to an exemplary embodiment, the prime mover **52** is configured to provide power to drive the rear tractive assembly **56** and/or the front tractive assembly **58** (e.g., to provide front-wheel drive, rear-wheel drive, four-wheel drive, and/or all-wheel drive operations). In some embodiments, the driveline **50** includes a transmission device (e.g., a gearbox, a continuous variable transmission (“CVT”), etc.) positioned between (a) the prime mover **52** and (b) the rear tractive assembly **56** and/or the front tractive assembly **58**. The rear tractive assembly **56** and/or the front tractive assembly **58** may include a drive shaft, a differential, and/or an axle. In some embodiments, the rear tractive assembly **56** and/or the front tractive assembly **58** include two axles or a tandem axle arrangement. In some embodiments, the rear tractive assembly **56** and/or the front tractive assembly **58** are steerable (e.g., using the steering wheel **42**). In some embodiments, both the rear tractive assembly **56** and the front tractive assembly **58** are fixed and not steerable (e.g., employ skid steer operations). By way of example, the driveline **50** may include a hydrostatic transmission that permits independent driving of the left and right sides of the driveline **50**.

[0033] In some embodiments, the driveline **50** includes a plurality of prime movers **52**. By way of example, the driveline **50** may include a first prime mover **52** that drives the rear tractive assembly **56** and a second prime mover **52** that drives the front tractive assembly **58**. By way of another example, the driveline **50** may include a first prime mover **52** that drives a first one of the front tractive elements, a second prime mover **52** that drives a second one of the front tractive elements, a third prime mover **52** that drives a first one of the rear tractive elements, and/or a fourth prime mover **52** that drives a second one of the rear tractive elements. By way of still another example, the driveline **50** may include a first prime mover **52** that drives the front tractive assembly **58**, a second prime mover **52** that drives a first one of the rear tractive elements, and a third prime mover **52** that drives a second one of the rear tractive elements. By way of yet another example, the driveline **50** may include a first prime mover **52** that drives the rear tractive assembly **56**, a second prime mover **52** that drives a first one of the front tractive elements, and a third prime mover **52** that drives a second one of the front tractive elements.

[0034] According to an exemplary embodiment, the suspension system **60** includes one or more suspension components (e.g., shocks, dampers, springs, etc.) positioned between the frame **12** and one or more components (e.g., tractive elements, axles, etc.) of the rear tractive assembly **56** and/or

the front tractive assembly **58**. In some embodiments, the mower **10** does not include the suspension system **60**.

[0035] According to an exemplary embodiment, the braking system **70** includes one or more braking components (e.g., disc brakes, drum brakes, in-board brakes, axle brakes, etc.) positioned to facilitate selectively braking one or more components of the driveline **50**. In some embodiments, the one or more braking components include (i) one or more front braking components positioned to facilitate braking one or more components of the front tractive assembly **58** (e.g., the front axle, the front tractive elements, etc.) and (ii) one or more rear braking components positioned to facilitate braking one or more components of the rear tractive assembly **56** (e.g., the rear axle, the rear tractive elements, etc.). In some embodiments, the one or more braking components include only the one or more front braking components. In some embodiments, the one or more braking components include only the one or more rear braking components. In some embodiments, the one or more front braking components include two front braking components, one positioned to facilitate braking each of the front tractive elements. In some embodiments, the one or more rear braking components include two rear braking components, one positioned to facilitate braking each of the rear tractive elements. In some embodiments, the driveline **50** is a hydrostatic transmission that performs braking by using hydraulic motors to oppose movement of the tractive elements.

[0036] As shown in FIG. 2, the control system **100** includes a controller **110**, one or more sensors **120**, and a communications interface **130** (e.g., located on the mower **10**). In some embodiments, the control system includes the remote system(s) **200**. In one embodiment, the controller **110** is configured to selectively engage, selectively disengage, control, or otherwise communicate with components of the mower **10**. According to an exemplary embodiment, the controller **110** is coupled to (e.g., communicably coupled to) components of the operator controls **40** (e.g., the steering wheel **42**, the traction pedal **44**, the operator interface **48**, etc.), components of the driveline **50** (e.g., the prime mover **52**), components of the braking system **70**, the sensors **120**, the communications interface **130**, and the remote system(s) **200**. By way of example, the controller **110** may send and receive signals (e.g., control signals, location signals, etc.) with the components of the operator controls **40**, the components of the driveline **50**, the components of the braking system **70**, the sensors **120**, the communications interface **130**, and/or the remote system(s) **200**.

[0037] The controller **110** may be implemented as a general-purpose processor, an application specific integrated circuit ("ASIC"), one or more field programmable gate arrays ("FPGAs"), a digital-signal-processor ("DSP"), circuits containing one or more processing components, circuitry for supporting a microprocessor, a group of processing components, or other suitable electronic processing components. According to the exemplary embodiment shown in FIG. 2, the controller **110** includes a processing circuit **112** and a memory **114**. The processing circuit **112** may include an ASIC, one or more FPGAs, a DSP, circuits containing one or more processing components, circuitry for supporting a microprocessor, a group of processing components, or other suitable electronic processing components. In some embodiments, the processing circuit **112** is configured to execute computer code stored in the memory **114** to facilitate the activities described herein. The memory **114** may be any volatile or non-volatile or non-transitory computer-readable storage medium capable of storing data or computer code relating to the activities described herein. According to an exemplary embodiment, the memory **114** includes computer code modules (e.g., executable code, object code, source code, script code, machine code, etc.) configured for execution by the processing circuit **112**. In some embodiments, the controller **110** may represent a collection of processing devices. In such cases, the processing circuit **112** represents the collective processors of the devices, and the memory **114** represents the collective storage devices of the devices.

[0038] The sensors **120** may include various sensors positioned about the mower **10** to acquire information or data regarding operation of the mower **10** and/or the location thereof. By way of example, the sensors **120** may include an accelerometer, a gyroscope, a compass, a position sensor

(e.g., a GPS sensor, etc.), suspension sensor(s), wheel sensors, an audio sensor or microphone, and/or other sensors to facilitate acquiring information or data regarding operation of the mower **10** and/or the location thereof. According to an exemplary embodiment, one or more of the sensors **120** are configured to facilitate detecting and obtaining vehicle telemetry data including position of the mower **10**, whether the mower **10** is moving, travel direction of the mower **10**, slope of the mower **10**, speed of the mower **10**, vibrations experienced by the mower **10**, sounds proximate the mower **10**, suspension travel of components of the suspension system **60**, and/or other vehicle telemetry data.

[0039] The communications interface **130** may be configured to facilitate wireless communications with the remote system(s) **200**. By way of example, the communications interface **130** may be configured to employ one or more types of wireless communications protocols including Bluetooth, Wi-Fi, radio, cellular, and/or other suitable wireless communications protocols.

[0040] The remote systems **200** may be or include an off-site server-based system that monitors various global positioning system (“GPS”) information and/or real-time kinematics (“RTK”) information (e.g., position/location, speed, direction of travel, geofence related information, etc.) and provides GPS data and/or RTK data based on the GPS information and/or RTK information to the controller **110** of the mower **10** through the communications interface **130**. The remote systems **200** may additionally or alternatively be or include an on-site system (e.g., in a club house of a golf course, on the golf course, etc.) that communicates with the mower **10** via the communications interface **130**. The on-site system may collect data from the mower **10** that may be used by the operators of the site (e.g., for advanced scheduling purposes, to identify persons braking course guidelines or rules, etc.). The on-site system may also function as an intermediary between the mower **10** and the off-site server-based system (e.g., if the communications interface **130** does not have long-range wireless communications capabilities). In some embodiments, the remote system(s) **200** include a processing circuit, a memory, and a communications interface similar to the control system **100**. In some embodiments (e.g., when the remote system(s) **200** are part of the control system **100**), the control system **100** includes a plurality of processing circuits **112** (e.g., a first processing circuit for the controller **110** of the mower **10** and a second processing circuit for the remote system(s) **200**) and a plurality of memories **114** (e.g., a first memory for the controller **110** of the mower **10** and a second memory for the remote system(s) **200**).

Mower Deck Arrangement

[0041] Referring to FIG. 1, the mower **10** includes a series of mower decks **80**, referred to generally herein as mower decks **80**. The mower decks **80** include a pair of front mower decks, shown as mower decks **80F**, a pair of rear mower decks, shown as mower decks **80R**, and a central mower deck, shown as mower deck **80C**. The mower decks **80F** are arranged symmetrically about a longitudinal centerline of the frame **12** such that one mower deck **80F** is positioned on a right side of the mower **10**, and another mower deck **80F** is positioned on left side of the mower **10**. The mower deck **80C** is positioned along the longitudinal centerline, between the mower decks **80F**. The mower decks **80R** are positioned rearward of the mower decks **80F** and the mower deck **80C**. The mower decks **80R** are arranged symmetrically about the longitudinal centerline such that one mower deck **80R** is positioned on a right side of the mower **10**, and another mower deck **80R** is positioned on left side of the mower **10**.

[0042] The mower **10** is selectively reconfigurable between (a) an operating, operational, or use configuration, shown in FIG. 1 and (b) a maintenance, service, or transport configuration, shown in FIG. 3. In the operating configuration, the mower decks **80** are coupled to the frame **12** and configured for use (e.g., cutting vegetation). In the service configuration, one or more of the mower decks **80** may be removed from the frame **12**. The service configuration may increase the ground clearance of the mower **10**, permitting the mower **10** to navigate onto surfaces that might otherwise contact the mower decks **80** (e.g., when driving up a ramp, when driving onto a trailer, etc.). Accordingly, the service configuration may facilitate shipping or transportation of the mower **10**.

The removal of the mower decks **80** in the service configuration may facilitate access to components and areas of the mower that might otherwise be obstructed by the mower decks **80**. Accordingly, the service configuration may facilitate performing maintenance or service procedures (e.g., cleaning, replacement of components, changing of fluids, etc.).

[0043] In some embodiments, the mower **10** is configurable into multiple different service configurations. By way of example, in a full or complete service configuration, all of the mower decks **80** may be removed from the frame **12**. By way of example, in a partial service configuration a first subset of the mower decks **80** may be removed from the frame **12** while a second subset of the mower decks **80** remain coupled to the frame **12**. A partial service configuration may be used, for example, when maintenance personnel require access to an area that is only obstructed by the first subset of the mower decks **80**.

[0044] Referring to FIGS. **1** and **4**, the construction of a mower deck **80F** is shown according to an exemplary embodiment. The mower decks **80R** and **80C** may have a similar configuration to the mower deck **80F**, as shown in FIGS. **12A** and **13**. Each mower deck **80** includes a base, deck, housing, frame, or enclosure, shown as housing **82**, and series of cutting elements **84** (e.g., reels, blades, flails, etc.) movably coupled to the housing **82**. The housing **82** may provide the structure to support the other components of the mower deck **80**. The housing **82** may open downward to expose the cutting elements **84** to vegetation below the housing **82**. Each mower deck **80** includes an interface, shown as post **86**, that extends upward from the housing **82**. The post **86** may be fixedly coupled to the housing **82** and facilitate coupling the housing **82** to the frame **12**.

[0045] Each mower deck **80** is coupled to a motor or actuator (e.g., an electric motor, a hydraulic motor, etc.), shown as mower motor **300**. Specifically, each mower motor **300** is removably coupled to the housing **82** of the corresponding mower deck **80** by a series of fasteners, shown as mounting bolts **302**. The mower motor **300** defines three apertures or passages (e.g., as shown in FIG. **9**), shown as fastener apertures **304**, through which the mounting bolts **302** extend. The fastener apertures **304** may be evenly spaced along the circumference of the mower motor **300**. The mounting bolts **302** may be in threaded engagement with the housing **82** to fixedly couple the mower motor **300** to the housing **82**. Additionally or alternatively, the mounting bolts **302** may be in threaded engagement with a secondary fastener, such as a nut, that secures the mower motor **300** to the housing **82**. The mower motor **300** may provide rotational mechanical energy (e.g., through an output shaft or motor shaft **306**, as shown in FIG. **10**) to drive movement (e.g., rotation, oscillation, etc.) of the cutting elements **84**. While driven by the mower motor **300**, the cutting elements **84** cut, crush, mulch, remove, or otherwise trim vegetation beneath the housing **82**. Operation of the mower motor **300** may be controlled by the control system **100**.

[0046] Each mower motor **300** is operatively coupled to the energy storage **54** and/or the control system **100** by a flexible member or tensile member (e.g., a wire, cable, or hose), shown as cable **310**. The cables **310** may transfer energy (e.g., in the form of electrical energy or pressurized hydraulic fluid, etc.) to the mower motors **300** to power operation of the mower motors **300**. A first end portion of each cable **310** may be coupled to the mower motor **300**, and an opposing second end portion of each cable **310** may be coupled to the frame **12**. The cables **310** may be flexible to permit movement of the mower decks **80** relative to the chassis **12**.

[0047] Each mower deck **80** is coupled to the frame **12** by a connecting shaft **320**. A first end portion or interface of the connecting shaft **320**, shown as receiver **322**, receives the post **86** of the mower deck **80**, pivotally coupling the mower deck **80** to the connecting shaft **320**. The receiver **322** defines a recess or passage that receives the post **86**, permitting rotation of the mower deck **80** relative to the connecting shaft **320** about a substantially vertical axis **324** that is centered about the post **86**. A second end portion or interface of the connecting shaft **320**, shown as shaft end **326**, extends away from the receiver **322** and is centered about a longitudinal axis **328**.

[0048] Referring to FIGS. **4-8**, each connecting shaft **320** is coupled to the frame **12** by a lift arm **340**. A first end portion or interface of the lift arm **340**, shown as receiver **342**, defines a passage

that extends along the longitudinal axis **328** and receives the shaft end **326** of the connecting shaft **320**, pivotally coupling the connecting shaft **320** to the lift arm **340**. The receiver **342** permits rotation of the connecting shaft **320** relative to the lift arm **340** about the longitudinal axis **328**. A second end portion or interface of the connecting shaft **320**, shown as lift arm shaft **344**, extends along and is centered about a longitudinal axis **346** that is offset from the longitudinal axis **328**. The lift arm shaft **344** is received by the frame **12**, pivotally coupling the lift arm **340** to the frame **12**. Accordingly, the lift arm **340** is rotatable relative to the frame **12** about the longitudinal axis **346**. A main portion of the lift arm **340**, shown as lift arm body **348**, extends between the receiver **342** and the lift arm shaft **344**.

[0049] The lift arm **340** includes a pair of mounting brackets or plates, shown as float plate **350** and float plate **352**, fixedly coupled to the lift arm body **348**. The lift arm body **348** extends between the float plate **350** and the float plate **352**, offsetting the float plates **350** and **352** such that the float plates **350** and **352** are positioned on opposite sides of the lift arm body **348**. The float plate **350** and the float plate **352** each define a slot or passage, shown as pin slot **354**, that extends along a length of the lift arm **340**. The pin slots **354** are aligned with one another. A pin or shaft, shown as float pin **356**, extends longitudinally through the pin slots **354**.

[0050] A linear actuator or lift actuator, shown as deck actuator **358**, has a first end coupled to the float pin **356** and a second end pivotally coupled to the frame **12**. The deck actuator **358** may extend or retract to vary a distance between the float pin **356** and the frame **12**. By way of example, the deck actuator **358** may include an electric linear actuator or a hydraulic cylinder. As shown in FIG. 2, the deck actuator **358** may be operatively coupled to the controller **110**. In other embodiments, the deck actuator **358** is another type of actuator, such as a lever that is manually operated by a user or a power take off from a prime mover (e.g., an engine, an electric motor, etc.) of the mower **10** (e.g., connected to the mower deck **80** by one or more belts).

[0051] A user may set a target height of the mower deck **80** (e.g., with the operator interface **48**). By way of example, the user may set the mower deck **80** to a relatively low cutting height to permit trimming vegetation. By way of another example, the user may set the mower deck **80** to a relatively high travel height to permit travel at elevated speeds without the mower deck **80** contacting the ground. The controller **110** may operate the deck actuator **358** to bring the corresponding mower deck **80** to the target height.

[0052] When the deck actuator **358** retracts, the float pin **356** contacts the proximal ends of the pin slots **354**, rotating the lift arm **340** upward. This upward rotation of the lift arm **340** raises the connecting shaft **320** and the mower deck **80**. Because the connecting shaft **320** is pivotable relative to the lift arm **340**, gravity acting on mower deck **80** may level the mower deck after this action. When the deck actuator **358** extends, the float pin **356** moves away from the frame **12**, and the lift arm is permitted to rotate downward. This downward rotation of the lift arm **340** lowers the connecting shaft **320** and the mower deck **80**. Because the connecting shaft **320** is pivotable relative to the lift arm **340**, gravity acting on mower deck **80** may level the mower deck after this action.

[0053] When the mower deck **80** encounters a ground surface having an upward change in elevation (e.g., a bump or hill), the ground surface may force the mower deck **80** upward. The arrangement of the lift arm **340** may permit the mower deck **80** to float freely upward to accommodate this upward force. Specifically, the lengths of the pin slots **354** may permit free movement of the float pin **356**. When the upward force is applied, the float pin **356** moves freely toward the distal ends of the pin slots **354**, permitting the mower deck **80** to move upward. When the upward force is reduced (e.g., the ground surface has a decrease in elevation), gravity draws the mower deck **80** downward and the float pin **356** moves back toward the proximal ends of the pin slots **354**.

[0054] Referring to FIGS. 3, 10, and 11, the lift arms **340** corresponding to the mower decks **80F** and **80R** are each configured to removably and fixedly couple to a corresponding mower motor **300** to support the mower motor **300** when the mower **10** is in the service configuration. When fixed to

the lift arm **340**, the mower motor **300** moves with the lift arm **340** relative to the chassis, and the corresponding cable **310** loops back toward the frame **12**.

[0055] As shown in FIGS. **5-7, 9**, and **10**, the float plate **352** includes a pair of vertical protrusions or motor mounting features, shown as tab **370** and tab **372**. The tabs **370** and **372** extend upward, above the pin slot **354**. As shown, the float plate **350** omits the tabs **370** and **372**, such that the float plate **352** extends further in an upward direction than (i.e., higher than) the float plate **350**. The tabs **370** and **372** are laterally offset by one another to form a gap, opening, recess, or space, shown as motor shaft recess **374**.

[0056] The tab **370** defines a first longitudinal passage or aperture, shown as fastener passage **380**. The tab **372** defines a second longitudinal passage or aperture, shown as fastener passage **382**, and a third longitudinal passage or aperture, shown as fastener passage **384**. The fastener passages **380** and **382** are positioned approximately the same distance from the pin slot **354** of the float plate **352**. The fastener passage **384** is positioned between the fastener passage **382** and the pin slot **354**, such that a distance from the pin slot **354** to the fastener passage **382** is greater than a distance from the pin slot **354** to the fastener passage **384**.

[0057] Each of the fastener passages **380, 382**, and **384** defines a female thread that is sized to threadedly engage a corresponding male thread of one of the mounting bolts **302**. As shown in FIGS. **9** and **10**, the mower motor **300** may be coupled to the float plate **352** by the mounting bolts **302**. A first distance between the fastener passage **380** and the fastener passage **382** may be approximately equal to a second distance between two of the fastener apertures **304**, such that the fastener passage **380** and the fastener passage **382** may each align with one of the fastener apertures **304**. In some embodiments, the fastener apertures **304** are uniformly spaced along the circumference of the mower motor **300**, such that the fastener passages **380** and **382** may be aligned with any two of the fastener apertures **304**.

[0058] To couple the mower motor **300** to the float plate **352**, a first mounting bolt **302** is inserted through one of the fastener apertures **304** and placed in threaded engagement with the fastener passage **380**. A second mounting bolt **302** is inserted through another of the fastener apertures **304** and placed in threaded engagement with the fastener passage **382**. When both mounting bolts **302** have been tightened, the mower motor **300** is fixed to the float plate **352**. Additionally, the motor shaft **306** extends through the motor shaft recess **374**, such that the motor shaft recess **374** provides clearance for the motor shaft **306**. As shown in FIG. **11**, a body of the mower motor **300** extends rearward from the float plate **352** and overhangs the float plate **350**. Accordingly, the float plate **350** being shorter than the float plate **352** provides clearance for the mower motor **300** to overhang the float plate **350**. In some embodiments, the mower motor **300** rests atop the float plate **350** to provide additional support to the mower motor **300**.

[0059] While more than two mounting bolts **302** may not be necessary to secure the mower motor **300** to the float plate **352** in the service configuration, the mower **10** may utilize three or more mounting bolts **302** (e.g., three mounting bolts **302** as shown in various figures herein) to couple the mower motor **300** to the housing **82** in the operating configuration. Accordingly, it may be desirable to secure these additional mounting bolts **302** to the lift arm **340** (e.g., so the mounting bolts **302** all remain with the mower **10** and are not lost). To facilitate this, the third mounting bolt **302** may be threaded into the fastener passage **384** and tightened, securing the mounting bolt **302** to the float plate **352**.

[0060] FIGS. **12A** and **12B** illustrate a mower deck **80R** and a lift arm **340** for use with the mower deck **80R**. The arrangement of the mower deck **80R** and corresponding lift arm **340** may be substantially similar to that of FIG. **4** except as otherwise specified. The lift arms **340** that support the mower decks **80R** are arranged with the float plates **350** and **352** inverted. By way of example, the float plate **352** supporting the mower deck **80F** in FIG. **4** is arranged forward of the corresponding float plate **350**, whereas the float plate **352** supporting the mower deck **80R** in FIG. **12A** is arranged rearward of the corresponding float plate **350**. The mower motors **300** overhang

the float plates **350**, such that this inverted arrangement of the float plates **350** and **352** also inverts the orientations of the mower motors **300** in the service configuration. By way of example, the body of the mower motor **300** for the mower deck **80F** extends rearward from the float plate **352** (as shown in FIG. **10**), and the body of the mower motor **300** for the mower deck **80R** extends rearward from the float plate **352** (as shown in FIG. **12B**).

[0061] Referring to FIGS. **13-17**, the mower **10** includes a mounting bracket or motor mount, shown as mounting plate **390**, that is fixedly coupled to the frame **12**. The mounting plate **390** is configured to removably and fixedly couple to the mower motor **300** corresponding to the mower deck **80C** to support the mower motor **300** when the mower **10** is in the service configuration. When fixed to the mounting plate **390**, the mower motor **300** is fixed to the frame **12**, and the corresponding cable **310** extends away from the loops back toward the frame **12**. The mounting plate **390** may be used instead of directly mounting the mower motor **300** of the mower deck **80C** to the corresponding float plate **352** for clearance or accessibility. In other embodiments, the mower motor **300** of the mower deck **80C** is directly coupled to the corresponding float plate **352**, and/or the mounting plate **390** is omitted.

[0062] The operator platform **34** includes a flange or plate, shown as flange **392**, that is fixedly coupled to the rest of the operator platform and extends below the rest of the operator platform **34**. The flange **392** extends vertically and longitudinally. The flange **392** is positioned along a left side of the operator platform **34** and extends downward from the top portion of the operator platform **34**. The flange **392** extends longitudinally along the operator platform. A pair of fasteners, shown as bolts **394**, fixedly couple the mounting plate **390** to the flange **392**. When coupled to the flange **392**, the mounting plate **390** extends below the flange **392**.

[0063] The mounting plate **390** includes a pair of vertical protrusions or motor mounting features, shown as tab **400** and tab **402**. The tabs **400** and **402** extend downward, below the flange **392** and the operator platform **34**. The tabs **400** and **402** are longitudinally offset from one another to form a gap, opening, recess, or space, shown as motor shaft recess **404**.

[0064] The tab **400** defines a first lateral passage or aperture, shown as fastener passage **410**. The tab **402** defines a second lateral passage or aperture, shown as fastener passage **412**, and a third lateral passage or aperture, shown as fastener passage **414**. The fastener passages **410** and **412** are positioned approximately the same distance from the flange **392**. The fastener passage **410**, the fastener passage **412**, and the fastener passage **414** are all positioned approximately along a longitudinal line (e.g., at approximately the same vertical position). The fastener passage **412** is positioned between the fastener passage **410** and the fastener passage **414**, such that a distance from the fastener passage **410** to the fastener passage **412** is greater than a distance from the fastener passage **410** to the fastener passage **414**.

[0065] Each of the fastener passages **410**, **412**, and **414** defines a female thread that is sized to threadedly engage a corresponding male thread of one of the mounting bolts **302**. As shown in FIGS. **15-17**, the mower motor **300** may be coupled to the mounting plate **390** by the mounting bolts **302**. A first distance between the fastener passage **410** and the fastener passage **412** may be approximately equal to a second distance between two of the fastener apertures **304**, such that the fastener passage **410** and the fastener passage **412** may each align with one of the fastener apertures **304**. In some embodiments, the fastener apertures **304** are uniformly spaced along the circumference of the mower motor **300**, such that the fastener passages **410** and **412** may be aligned with any two of the fastener apertures **304**.

[0066] To couple the mower motor **300** to the mounting plate **390**, a first mounting bolt **302** is inserted through one of the fastener apertures **304** and placed in threaded engagement with the fastener passage **410**. A second mounting bolt **302** is inserted through another of the fastener apertures **304** and placed in threaded engagement with the fastener passage **412**. When both mounting bolts **302** have been tightened, the mower motor **300** is fixed to the mounting plate **390**, and thus fixed to the frame **12**. Additionally, the motor shaft **306** extends through the motor shaft

recess **404**, such that the motor shaft recess **404** provides clearance for the motor shaft **306**.

[0067] While more than two mounting bolts **302** may not be necessary to secure the mower motor **300** to the mounting plate **390** in the service configuration, the mower **10** may utilize three or more mounting bolts **302** (e.g., three mounting bolts **302** as shown in various figures herein) to couple the mower motor **300** to the housing **82** of the mower deck **80C** in the operating configuration. Accordingly, it may be desirable to secure these additional mounting bolts **302** to the frame **12** (e.g., so the mounting bolts **302** all remain with the mower **10** and are not lost). To facilitate this, the third mounting bolt **302** may be threaded into the fastener passage **414** and tightened, securing the mounting bolt **302** to the mounting plate **390**.

[0068] In some embodiments, some or all of the mounting bolts **302** may be the same size (e.g., length, diameter, and/or thread pitch), such that the mounting bolts **302** are interchangeable with one another. This may be desirable to reduce complexity and increase flexibility during the assembly process. By way of example, a user may utilize a given mounting bolt **302** to do any of (a) secure a mower motor **300** to a lift arm **340**, (b) secure a mower motor **300** to a mounting plate **390**, or (b) store the mounting bolt **302** in fastener passage **384** or a fastener passage **414**. The mounting bolts **302** may be interchangeable between mower decks **80**.

[0069] Referring generally to the figures, a user may reconfigure the mower **10** from operating configuration to the service configuration by (a) disconnecting or removing the mower motors **300** from the corresponding mower decks **80**, (b) disconnecting or removing the mower decks **80** from the frame **12**, and (c) coupling the mower motors **300** to the respective mounting plates (e.g., the float plates **352** or the mounting plate **390**). The reverse of this process may be followed to reconfigure the mower **10** from the service configuration to the operating configuration.

[0070] To disconnect the mower motors **300** from the mower decks **80**, the mounting bolts **302** for each mower motor **300** may be loosened and removed (e.g., using a wrench). At this point, the mower motors **300** may hang freely from the frame **12** by the cables **310**. The user may disconnect the mower decks **80** from the frame **12** by removing the posts **86** from the corresponding receivers **322**. By way of example, a post **86** may be released from a receiver **322** by removing a retaining pin.

[0071] A mower motor **300** may be coupled to a float plate **352** by aligning two of the fastener apertures **304** with the fastener passage **380** and the fastener passage **382**. A first mounting bolt **302** may be inserted through a fastener aperture **304** and into the fastener passage **380**, and a second mounting bolt **302** may be inserted through a fastener aperture **304** and into the fastener passage **380**. The first and second mounting bolts **302** may be tightened (e.g., using a wrench) to fix the mower motor **300** to the float plate **352**. A third mounting bolt **302** may be inserted into the fastener passage **384** and tightened to fix the third mounting bolt **302** to the float plate **352**. A similar process may be followed to couple a mower motor **300** to a mounting plate **390**.

[0072] As utilized herein with respect to numerical ranges, the terms “approximately,” “about,” “substantially,” and similar terms generally mean $\pm 10\%$ of the disclosed values, unless specified otherwise. As utilized herein with respect to structural features (e.g., to describe shape, size, orientation, direction, relative position, etc.), the terms “approximately,” “about,” “substantially,” and similar terms are meant to cover minor variations in structure that may result from, for example, the manufacturing or assembly process and are intended to have a broad meaning in harmony with the common and accepted usage by those of ordinary skill in the art to which the subject matter of this disclosure pertains. Accordingly, these terms should be interpreted as indicating that insubstantial or inconsequential modifications or alterations of the subject matter described and claimed are considered to be within the scope of the disclosure as recited in the appended claims.

[0073] It should be noted that the term “exemplary” and variations thereof, as used herein to describe various embodiments, are intended to indicate that such embodiments are possible examples, representations, or illustrations of possible embodiments (and such terms are not

intended to connote that such embodiments are necessarily extraordinary or superlative examples).

[0074] The term “coupled” and variations thereof, as used herein, means the joining of two members directly or indirectly to one another. Such joining may be stationary (e.g., permanent or fixed) or moveable (e.g., removable or releasable). Such joining may be achieved with the two members coupled directly to each other, with the two members coupled to each other using a separate intervening member and any additional intermediate members coupled with one another, or with the two members coupled to each other using an intervening member that is integrally formed as a single unitary body with one of the two members. If “coupled” or variations thereof are modified by an additional term (e.g., directly coupled), the generic definition of “coupled” provided above is modified by the plain language meaning of the additional term (e.g., “directly coupled” means the joining of two members without any separate intervening member), resulting in a narrower definition than the generic definition of “coupled” provided above. Such coupling may be mechanical, electrical, or fluidic.

[0075] References herein to the positions of elements (e.g., “top,” “bottom,” “above,” “below”) are merely used to describe the orientation of various elements in the figures. It should be noted that the orientation of various elements may differ according to other exemplary embodiments, and that such variations are intended to be encompassed by the present disclosure.

[0076] The hardware and data processing components used to implement the various processes, operations, illustrative logics, logical blocks, modules, and circuits described in connection with the embodiments disclosed herein may be implemented or performed with a general purpose single-or multi-chip processor, a digital signal processor (DSP), an application specific integrated circuit (ASIC), a field programmable gate array (FPGA), or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general purpose processor may be a microprocessor, or any conventional processor, controller, microcontroller, or state machine. A processor also may be implemented as a combination of computing devices, such as a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration. In some embodiments, particular processes and methods may be performed by circuitry that is specific to a given function. The memory (e.g., memory, memory unit, storage device) may include one or more devices (e.g., RAM, ROM, Flash memory, hard disk storage) for storing data and/or computer code for completing or facilitating the various processes, layers and modules described in the present disclosure. The memory may be or include volatile memory or non-volatile memory, and may include database components, object code components, script components, or any other type of information structure for supporting the various activities and information structures described in the present disclosure. According to an exemplary embodiment, the memory is communicably connected to the processor via a processing circuit and includes computer code for executing (e.g., by the processing circuit or the processor) the one or more processes described herein.

[0077] The present disclosure contemplates methods, systems, and program products on any machine-readable media for accomplishing various operations. The embodiments of the present disclosure may be implemented using existing computer processors, or by a special purpose computer processor for an appropriate system, incorporated for this or another purpose, or by a hardwired system. Embodiments within the scope of the present disclosure include program products comprising machine-readable media for carrying or having machine-executable instructions or data structures stored thereon. Such machine-readable media can be any available media that can be accessed by a general purpose or special purpose computer or other machine with a processor. By way of example, such machine-readable media can comprise RAM, ROM, EPROM, EEPROM, or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to carry or store desired program code in the form of machine-executable instructions or data structures and which can be accessed by a general

purpose or special purpose computer or other machine with a processor. Combinations of the above are also included within the scope of machine-readable media. Machine-executable instructions include, for example, instructions and data which cause a general purpose computer, special purpose computer, or special purpose processing machines to perform a certain function or group of functions.

[0078] Although the figures and description may illustrate a specific order of method steps, the order of such steps may differ from what is depicted and described, unless specified differently above. Also, two or more steps may be performed concurrently or with partial concurrence, unless specified differently above. Such variation may depend, for example, on the software and hardware systems chosen and on designer choice. All such variations are within the scope of the disclosure. Likewise, software implementations of the described methods could be accomplished with standard programming techniques with rule-based logic and other logic to accomplish the various connection steps, processing steps, comparison steps, and decision steps.

[0079] It is important to note that the construction and arrangement of the mower **10** and the systems and components thereof (e.g., the body **20**, the operator controls **40**, the driveline **50**, the suspension system **60**, the braking system **70**, the control system **100**, etc.) as shown in the various exemplary embodiments is illustrative only. Additionally, any element disclosed in one embodiment may be incorporated or utilized with any other embodiment disclosed herein. By way of example, the mower deck **80F** shown in FIG. **4** may be interchanged with the mower deck **80R** shown in FIG. **12A**.

Claims

1. A mower, comprising: a chassis; a tractive element coupled to the chassis; a cutting unit coupled to the chassis and including a cutting element; a mounting bracket coupled to the chassis; and a motor configured to drive the cutting element, wherein the motor is configured to be removably coupled to the cutting unit and configured to be removably coupled to the mounting bracket, wherein the mower is reconfigurable between (a) an operating configuration in which the motor is removed from the mounting bracket and coupled to the cutting unit and (b) a service or transportation configuration in which the motor is removed from the cutting unit and coupled to the mounting bracket.
2. The mower of claim 1, wherein the mounting bracket is pivotally coupled to the chassis such that the motor is repositionable relative to the chassis while in the service or transportation configuration.
3. The mower of claim 1, further comprising a lift arm coupling the cutting unit to the chassis and including the mounting bracket, wherein the lift arm is pivotally coupled to the chassis.
4. The mower of claim 3, further comprising a pin coupled to the chassis, wherein the mounting bracket defines a slot that receives the pin, and wherein the pin is configured to move along the slot in response to an upward movement of the cutting unit.
5. The mower of claim 4, further comprising a lift actuator coupled to the pin and the chassis and configured to move the pin relative to the chassis to raise the cutting unit.
6. The mower of claim 4, wherein the slot is a first slot, wherein the lift arm further includes a float bracket fixedly coupled to the mounting bracket and defining a second slot that receives the pin, wherein the motor overhangs the float bracket when the mower is in the service or transportation configuration.
7. The mower of claim 1, wherein the mounting bracket is fixedly coupled to the chassis.
8. The mower of claim 7, wherein the chassis includes an operator platform configured to support an operator, wherein the mounting bracket is directly coupled to the operator platform and extends below the operator platform.
9. The mower of claim 1, wherein the motor defines a fastener aperture and the mounting bracket

defines a fastener passage, further comprising a fastener configured to extend through the fastener aperture and into the fastener passage to removably couple the motor to the mounting bracket.

10. The mower of claim 9, wherein the fastener aperture is a first fastener aperture, the fastener passage is a first fastener passage, and the fastener is a first fastener, and wherein the motor defines a second fastener aperture and the mounting bracket defines a second fastener passage, further comprising a second fastener that is configured to extend through the second fastener aperture and into the second fastener passage to removably couple the motor to the mounting bracket.

11. The mower of claim 10, wherein the motor includes a motor shaft, wherein the mounting bracket includes a first tab that defines the first fastener passage and a second tab that defines the second fastener passage, and wherein the motor shaft extends through an opening between the first tab and the second tab.

12. The mower of claim 9, wherein the fastener is configured to extend through the fastener aperture and engage the cutting unit to removably couple the motor to the cutting unit.

13. A mower, comprising: a chassis; a tractive element coupled to the chassis; a cutting unit coupled to the chassis and including a cutting element; an actuator configured to raise the cutting unit relative to the chassis; a mounting bracket coupled to the chassis, the mounting bracket defining a first fastener passage and a second fastener passage; a motor configured to drive the cutting element, the motor defining a first fastener aperture and a second fastener aperture; and a first fastener and a second fastener, wherein the mower is reconfigurable between: a first configuration in which (a) the first fastener extends through the first fastener aperture and engages the cutting unit and (b) and the second fastener extends through the second fastener aperture and engages the cutting unit; and a second configuration in which (a) the first fastener extends through the first fastener aperture and into the first fastener passage to couple the motor to the mounting bracket and (b) the second fastener extends into the second fastener passage without extending through the second fastener aperture.

14. The mower of claim 13, wherein the mounting bracket is fixedly coupled to the chassis.

15. The mower of claim 13, wherein the mounting bracket is pivotally coupled to the chassis.

16. The mower of claim 13, wherein the first fastener passage is threaded to engage the first fastener, and wherein the second fastener passage is threaded to engage the second fastener passage.

17. The mower of claim 13, wherein the mounting bracket defines a third fastener passage and the motor defines a third fastener aperture, further comprising a third fastener, wherein the third fastener extends through the third fastener aperture and engages the cutting unit in the first configuration, and wherein the third fastener extends through the third fastener aperture and into the third fastener passage to couple the motor to the mounting bracket in the second configuration.

18. The mower of claim 13, wherein the cutting unit is removed from the chassis in the second configuration.

19. A mower, comprising: a chassis; a tractive element coupled to the chassis; a first cutting unit configured to be driven by a first motor; a second cutting unit configured to be driven by a second motor; a first mounting bracket pivotally coupled to the chassis; and a second mounting bracket fixedly coupled to the chassis, wherein the mower is reconfigurable between: a first configuration in which (a) the first motor is coupled to the first cutting unit and (b) the second motor is coupled to the second cutting unit; and a second configuration in which (a) the first motor is disconnected from the first cutting unit and coupled to the first mounting bracket and (b) the second motor is disconnected from the second cutting unit and coupled to the second mounting bracket.

20. The mower of claim 19, wherein the first cutting unit and the second cutting unit are disconnected from the chassis in the second configuration.
